



JAGANNATH UNIVERSITY

Department of Computer Science & Engineering

Course: Advanced Cloud Computing

Summer 2026 MSc Professional

AmazonAICloud: Proactive Resource Allocation Using Amazon Chronos
Based Time Series Model for Sustainable Cloud Computing

30 Marks Exam Questions & Answers

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Important: The questions in this book are intentionally generic, following the style of typical ACC final examination questions. The AmazonAICloud paper is used as the source of the topics, concepts, methods, equations, architecture, and results.

30-Mark Model Examination + Full Answers

Exam-oriented, structured, and easy to memorize

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1 How the Questions Are Designed

The questions below are not written as direct research-paper questions. Instead, they convert the important technical points of the AmazonAICloud study into the type of generic questions commonly used in Advanced Cloud Computing examinations.

Source focus: Cloud resource management, over/under-provisioning, reactive and proactive allocation, time-series forecasting, Amazon Chronos, Bitbrains workload traces, KPSS stationarity testing, forecasting metrics, cloud experimentation, model comparison, and sustainable cloud computing.

Exam pattern: Definitions → explanation → comparison → diagram → equation/result analysis.

2 30-Mark Generic Model Question

Question 1

- a. What is cloud computing? Briefly explain the major objectives of cloud resource management. [3 Marks]
- b. What is over-provisioning and under-provisioning? Explain their effects on cloud applications. [4 Marks]
- c. What do you mean by proactive resource allocation? Compare reactive and proactive resource allocation. [5 Marks]
- d. What is time-series forecasting? Explain the basic workflow of using historical CPU utilization for future resource prediction. [4 Marks]
- e. What is Amazon Chronos? Describe the main steps of its forecasting process with a suitable diagram. [6 Marks]
- f. Define MAE, MSE and MAPE with their equations. [4 Marks]
- g. Write the major findings of the forecasting experiment and explain how proactive resource allocation can support sustainable cloud computing. [4 Marks]

Total = 30 Marks

3 Question 1(a): Cloud Computing and Resource Management

What is cloud computing? Briefly explain the major objectives of cloud resource management.

Answer:

Cloud computing is a computing approach in which computing resources and services are provided through a network so that applications can obtain the required infrastructure and processing capacity when needed.

In the context of the AmazonAICloud study, an important responsibility of cloud computing is to manage resources according to changing workload demand.

Major objectives of cloud resource management:

1. Efficient resource utilization: Allocate resources according to application demand.
2. Application performance: Avoid insufficient resources that may degrade application performance and QoS.
3. Avoid resource wastage: Prevent allocating more resources than required.
4. Dynamic resource allocation: Adjust provisioning and de-provisioning as workload changes.
5. Cost and energy efficiency: Reduce unnecessary resource usage and support more sustainable cloud operation.

In short: Cloud resource management attempts to maintain a balance between performance and resource efficiency.

4 Question 1(b): Over-Provisioning and Under-Provisioning

What is over-provisioning and under-provisioning? Explain their effects on cloud applications.

Answer:

Cloud workloads are dynamic, so CPU and memory requirements can change over time. If resources are not matched with actual demand, two major problems can occur.

1. Over-Provisioning

Over-provisioning occurs when more cloud resources are allocated than the application actually requires.

Effects:

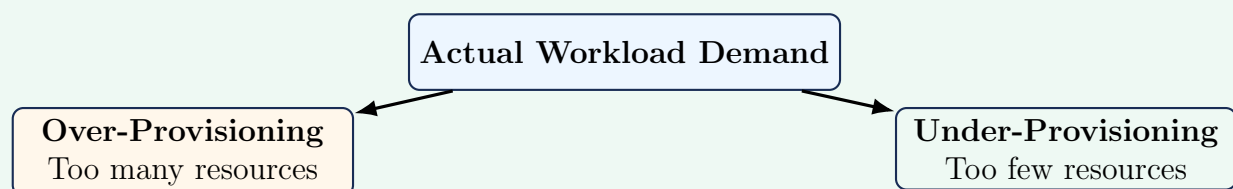
- Resources remain underutilized.
- Computing capacity is wasted.
- Operational cost can increase.
- Unnecessary resource and energy consumption may occur.

2. Under-Provisioning

Under-provisioning occurs when fewer resources are allocated than the application requires.

Effects:

- Application performance may degrade.
- Response quality/QoS may be affected.
- Sudden workload increases may cause insufficient capacity.
- Resource allocation decisions may become inefficient.



5 Question 1(c): Reactive and Proactive Resource Allocation

What do you mean by proactive resource allocation? Compare reactive and proactive resource allocation.

Answer:

Reactive resource allocation responds to a workload change after the change has been observed. A typical approach monitors resource utilization, detects a threshold condition, and then scales resources.

Observe Workload → Detect Threshold → Scale Resources

Proactive resource allocation predicts future workload demand before the expected change occurs and prepares resources in advance.

Historical Workload → Forecast Future Demand → Prepare Resources

Comparison

Aspect	Reactive Allocation	Proactive Allocation
Decision basis	Current/observed workload	Predicted future workload
Timing	After workload change	Before expected workload change
Typical mechanism	Monitoring and threshold	Time-series forecasting
Main advantage	Simple response to current state	Earlier preparation for future demand
Main risk	May react after a sudden spike	Depends on forecast accuracy

Conclusion: The AmazonAICloud approach uses forecasting to move from reactive resource management toward proactive resource allocation.

6 Question 1(d): Time-Series Forecasting

What is time-series forecasting? Explain the basic workflow of using historical CPU utilization for future resource prediction.

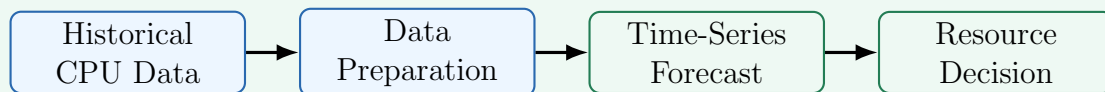
Answer:

Time-series forecasting is the process of using historical observations collected over time to estimate future values.

In the AmazonAICloud experiment, historical CPU utilization is used as the input and future CPU utilization is forecast.

Basic workflow:

1. Collect historical workload data: Obtain CPU utilization observations from cloud VM traces.
2. Prepare the time series: Organize and analyze the workload observations.
3. Apply forecasting model: Give the historical values to Amazon Chronos.
4. Generate forecast: Predict future CPU utilization.
5. Use forecast for management: The predicted workload provides information that can support proactive provisioning and de-provisioning decisions.



7 Question 1(e): Amazon Chronos

What is Amazon Chronos? Describe the main steps of its forecasting process with a suitable diagram.

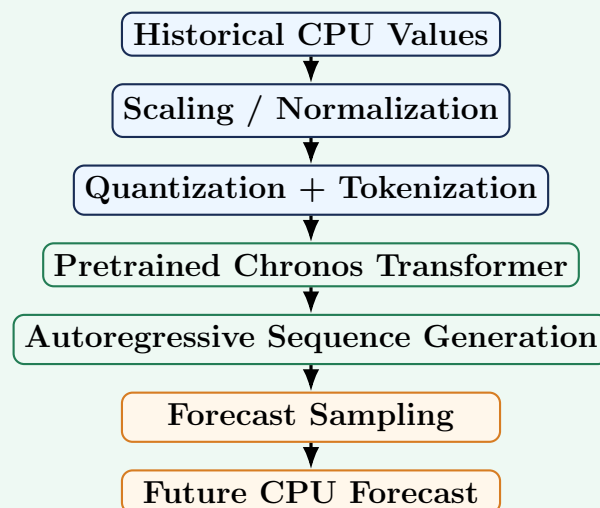
Answer: Amazon Chronos is a pretrained, transformer-based time-series forecasting model developed by Amazon AI Labs. The AmazonAICloud study uses the Chronos Tiny variant because of its relatively low computational requirements.

The model transforms numerical time-series observations into a discrete representation and generates future values autoregressively.

Main Steps

1. Input: Historical CPU utilization values are provided to the model.
2. Scaling: The context is normalized/scaled before tokenization.
3. Quantization/Tokenization: Numerical values are transformed into a discrete representation.
4. Transformer model: The sequence is processed by the pretrained forecasting model.
5. Autoregressive generation: Future sequence tokens are generated sequentially.
6. Sampling: Multiple forecast samples can be generated.
7. Output transformation: Predictions are converted back into CPU utilization values.

Chronos Forecasting Diagram



Study configuration: Chronos Tiny was configured with 30 forecast samples and a prediction length of 30 time steps.

Easy sequence to memorize:

Input → Scale → Tokenize → Chronos → Generate → Sample → Forecast

8 Question 1(f): MAE, MSE and MAPE

Define MAE, MSE and MAPE with their equations.

Let y_i be the actual value, $\hat{y}_i$ be the predicted value, and n be the number of observations.

1. Mean Absolute Error (MAE)

MAE measures the average magnitude of the prediction errors.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Interpretation: Lower MAE means lower average prediction error.

2. Mean Squared Error (MSE)

MSE calculates the average squared prediction error and therefore penalizes larger errors more strongly.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Interpretation: Lower MSE indicates fewer or smaller large prediction errors.

3. Mean Absolute Percentage Error (MAPE)

MAPE expresses the average absolute prediction error as a percentage of the actual value.

$$MAPE = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

Interpretation: Lower MAPE indicates better percentage-based forecasting accuracy.

MAE = Absolute Error MSE = Squared Error MAPE =
Percentage Error
Lower value = Better forecasting accuracy

9 Question 1(g): Results & Sustainable Cloud Computing

Write the major findings of the forecasting experiment and explain how proactive resource allocation can support sustainable cloud computing.

Answer:

The study compared ARIMA, LSTM, DeepAR and Amazon Chronos using forecasting error and computational-time measures.

Reported Forecasting Results

Model	MAE	MSE	MAPE
ARIMA	13.825	511.459	19.616
LSTM	8.217	1728.004	24.550
DeepAR	15.852	230.884	7.550
Amazon Chronos	6.332	91.390	6.450

Major findings:

- Amazon Chronos achieved the lowest reported MAE, MSE and MAPE among the compared models.
- ARIMA was the fastest model but produced higher forecasting errors.
- LSTM obtained a lower MAE than ARIMA but had the highest reported MSE and MAPE in the experiment.
- DeepAR had comparatively strong MSE and MAPE but required the longest reported training time.
- Chronos required 324 seconds of training, compared with 562 seconds for DeepAR, representing a reported reduction of about 42.35%.

Sustainability

Accurate workload forecasting can provide cloud resource managers with earlier information about future demand. This can help resources be provisioned or de-provisioned closer to expected demand.

Thus:

Better Forecast → Better Allocation → Less Unnecessary Resource Use → Cost Efficiency

Therefore, proactive resource allocation is presented as a pathway toward more efficient and sustainable cloud computing.

Important limitation: The study positions sustainability as a goal and implication; it does not directly measure energy or carbon reduction in the reported experiment.

10 Additional Paper Based Questions from the Amazon-AICloud0 Paper

The following questions are useful alternatives because they follow the same generic ACC examination style while remaining directly connected to the research-paper topics.

10.1 Question 1: Cloud Workload Dataset

- a. What is a cloud workload trace?
- b. Name the dataset used in the study and mention its major characteristics.
- c. What resource variables are recorded in the dataset? Explain why these variables are useful for cloud resource management.

Answer:

a. Cloud workload trace:

A cloud workload trace is a historical record of resource or workload measurements collected from cloud computing systems over time.

b. Dataset:

The study uses the Bitbrains GWA-T-12 cloud VM trace dataset. It contains performance metrics from approximately 1,750 virtual machines over the period 1 July 2013 to 1 October 2013.

c. Resource variables:

The dataset contains:

- CPU usage
- CPU provisioned
- CPU cores
- Memory usage
- Memory provisioned

These variables help characterize changing cloud workload demand and allow comparison between resources actually used and resources provisioned.

10.2 Question 2: Stationarity and KPSS Test

- a. What is a stationary time series?
- b. What is the KPSS test? Why was it applied in the study?
- c. Interpret the reported KPSS result.

Answer:**a. Stationary time series:**

A stationary time series has statistical properties that remain stable over time according to the stationarity assumption being tested.

b. KPSS test:

The Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test is a statistical test used to examine stationarity. In the study, it was used in addition to the earlier ADF-based analysis to further examine the stationarity of the Bitbrains workload data.

c. Reported result:

$$\text{KPSS statistic} = 0.432$$

$$p\text{-value} = 0.01$$

$$\text{Lags} = 27$$

The study treats the workload as non-stationary, because the reported p -value is below 0.05 and the statistic is above the reported critical values.

Exam line: The KPSS result supports the need to consider forecasting methods suitable for changing and non-stationary cloud workload behavior.

10.3 Question 3: Experimental Environment

- a. Which cloud platform/environment was used for the experiment?
- b. What was the forecasting input and output?
- c. Name the models compared in the experiment.

Answer:

a. Experimental environment:

The experiment was conducted using an AWS SageMaker Jupyter Notebook with the ml.t3.large CPU instance.

b. Forecasting task:

- Input: Historical CPU utilization.
- Output: Next 30 time steps.
- Forecast samples: 30.

c. Compared models:

1. ARIMA
2. LSTM
3. DeepAR
4. Amazon Chronos Tiny

10.4 Question 4: Model Comparison

Compare ARIMA, LSTM, DeepAR and Amazon Chronos based on the reported forecasting results.

Answer:

Model	MAE	MSE	MAPE	Train (s)	Inference (s)
ARIMA	13.825	511.459	19.616	0.03	0.012
LSTM	8.217	1728.004	24.550	40.8	0.6
DeepAR	15.852	230.884	7.550	562	1.7
Chronos	6.332	91.390	6.450	324	1.27

Interpretation:

- Chronos: Best overall forecasting accuracy because it has the lowest reported MAE, MSE and MAPE.
- ARIMA: Fastest in training and inference, but with higher forecasting errors.
- LSTM: Competitive MAE but highest reported MSE and MAPE in this experiment.
- DeepAR: Strong MSE and MAPE results but highest training time.

Conclusion: The reported results present Chronos as the best overall balance of forecast accuracy and computational practicality among the evaluated models.

10.5 Question 5: Architecture and Workflow

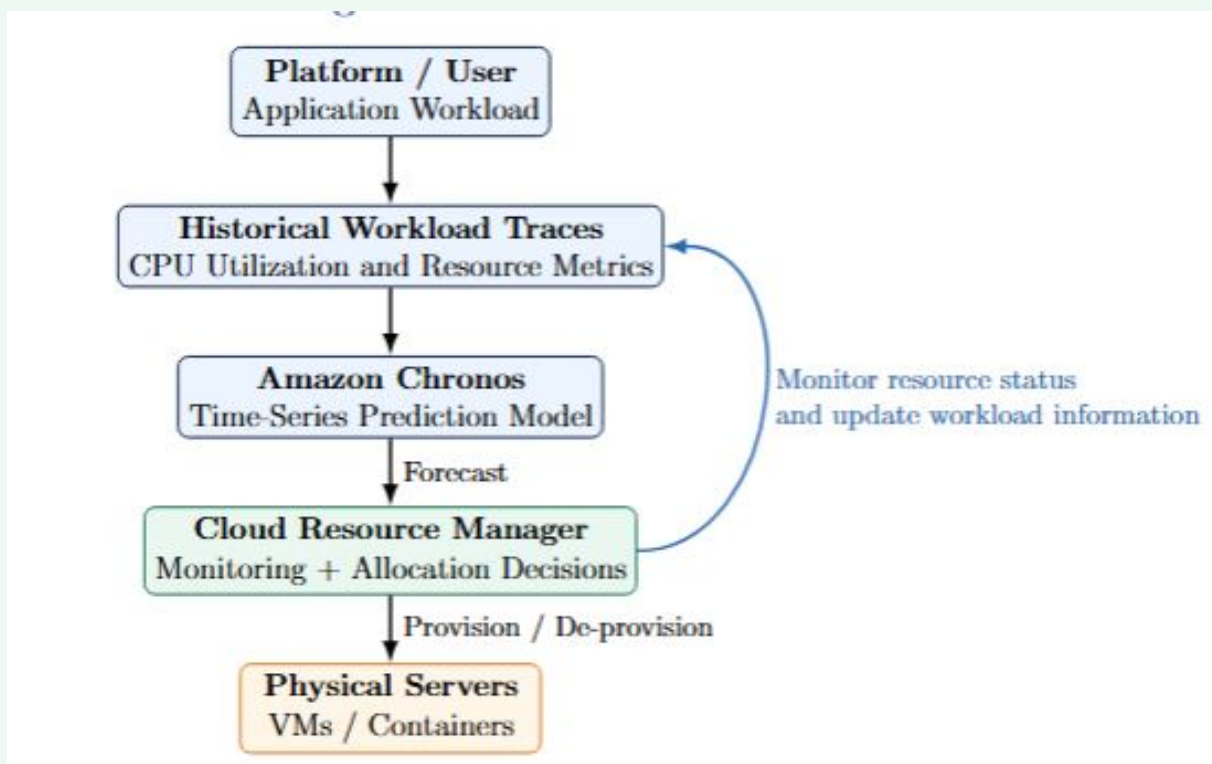
With a suitable diagram, explain the architecture of a cloud-based proactive resource allocation system.

Answer:

A proactive cloud resource allocation system uses historical workload information to predict future demand and then uses the prediction to support resource management decisions.

Main components:

1. Cloud Workload: Generates VM/application workload and resource measurements.
2. Data Processing: Collects and prepares workload history for forecasting.
3. Time-Series Prediction Model: Forecasts future resource demand. In the study, this role is performed by Amazon Chronos.
4. Cloud Resource Manager: Uses the forecast to support provisioning and de-provisioning decisions.
5. Cloud Infrastructure: Contains VMs, servers and applications where resources are actually allocated.
6. Feedback Loop: Actual utilization becomes new historical input, allowing forecasts to be repeatedly updated.



Workflow:

Workload H. → Processing → Forecast → Resource Decision → Infrastructure → Feedback

10.6 Question 6: Sustainable Cloud Computing

What is sustainable cloud computing? Explain how intelligent resource prediction can contribute to sustainability.

Answer:

Sustainable cloud computing aims to operate cloud computing systems efficiently while reducing unnecessary resource use and its associated environmental impact.

The AmazonAICloud study connects accurate forecasting with more efficient resource provisioning.

Contribution of prediction:

1. Predict future resource demand.
2. Avoid allocating excessive resources when demand is low.
3. Reduce the possibility of insufficient resources during expected demand increases.
4. Support more appropriate provisioning and de-provisioning.
5. Potentially reduce unnecessary operational resource and energy consumption.

Forecasting → Efficient Allo. → Less Resource Wastage → Potential Sustainability Benefit

Important: The paper presents energy and carbon optimization as an important future research direction; direct energy/carbon reduction was not measured in the reported experiment.

10.7 Question 7: Limitations and Future Work

What are the limitations and future research directions of the proposed approach?

Answer:

Important limitations/cautions:

- The reported experiment mainly focuses on forecasting performance.
- Energy and carbon reduction were not directly measured.
- Accurate forecasts do not automatically guarantee optimal resource allocation.
- Production deployment requires appropriate scaling policies and live operational validation.

Future research directions:

1. Investigate additional hybrid machine-learning models.
2. Integrate Chronos with real-time cloud performance metrics.
3. Explore larger Chronos variants such as Base and Large.
4. Use generative AI to simulate diverse resource-usage patterns.
5. Directly investigate energy consumption and carbon-emission optimization.
6. Explore explainable AI for transparency and trustworthiness.
7. Consider SLA-aware resource-management decisions.

11 Most Likely Generic Questions to Practice

1. What is cloud resource management? Explain its objectives.
2. What is over-provisioning? Explain its disadvantages.
3. What is under-provisioning? Explain its effect on QoS.
4. Compare reactive and proactive resource allocation.
5. What is workload forecasting? Explain its importance in cloud computing.
6. What is time-series forecasting?
7. What is Amazon Chronos?
8. Explain the working procedure of a pretrained time-series forecasting model.
9. Draw and explain a proactive cloud resource allocation architecture.
10. What is a cloud workload trace?
11. Describe the Bitbrains GWA-T-12 dataset.
12. What is stationarity in a time series?
13. What is the KPSS test? Why is it useful?
14. Define MAE, MSE and MAPE with equations.
15. Compare ARIMA, LSTM, DeepAR and Chronos.
16. Explain the experimental setup of a cloud forecasting experiment.
17. What is sustainable cloud computing?
18. Explain how workload prediction can reduce resource wastage.
19. Discuss the limitations of forecasting-based resource allocation.
20. Discuss future research directions for intelligent cloud resource management.

12 Final One-Page Revision Sheet

1. Main Problem

Dynamic cloud workloads cause over-provisioning and under-provisioning.

2. Traditional Approach

Reactive management:

Observe → Threshold → Scale

3. Proposed Idea

Proactive management:

History → Forecast → Prepare Resources

4. Forecasting Model

Amazon Chronos = pretrained transformer-based time-series model.

5. Dataset

Bitbrains GWA-T-12; 1,750 VMs; CPU and memory-related metrics; 2013-07-01 to 2013-10-01.

6. Stationarity

KPSS statistic = 0.432, $p = 0.01$, lags = 27. The study treats the workload as non-stationary.

7. Metrics

$$MAE = \frac{1}{n} \sum |y_i - \hat{y}_i|$$

$$MSE = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$$

$$MAPE = \frac{100}{n} \sum \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

8. Best Reported Model

Chronos: MAE = 6.332, MSE = 91.390, MAPE = 6.450.

9. Main Conclusion

Better forecasting can provide earlier information for resource decisions and can support more efficient cloud resource provisioning.

10. Sustainability Caution

The paper links better allocation with sustainability, but direct energy/carbon reduction was not measured in the reported experiment.

Exam Answer Formula

Definition → Key Points → Diagram/Equation → Example/Result → Conclusion